

Assignment 2

Maxwell Eichelberger, Joe Ferguson

Introduction

This assignment is about logistic regression with binary and multi-class prediction. It involves implementing sigmoid and softmax activation functions, gradient descent, and predictions for the binary classification, two-stage classification, and multi-class classification. The approach is as follows.

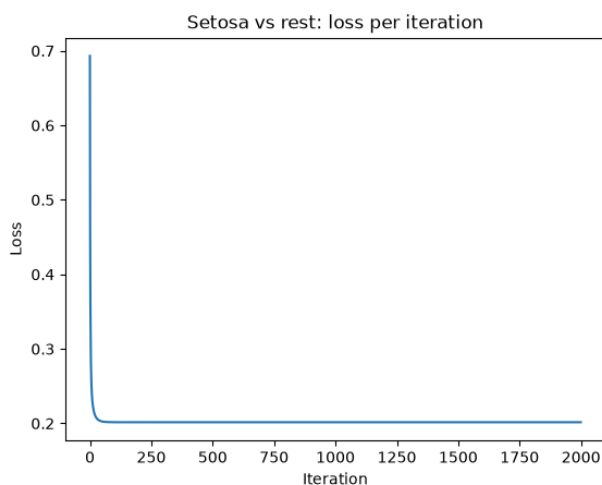
Part 1: Binary logistic regression

1.1 Implementation

I implemented and developed sigmoid, clipping the range of allowed z to $\text{abs}(500)$. I also implement the gradient descent and loss functions, making sure to clip the probabilities as well. I original had the loops, so I had to take the time to look at numpy documentation in order to vectorize the computations.

1.2 Train on the binary problem setosa (label 1) vs. the rest (label 0)

I encoded the labels to predict setosa flowers only, then fitted the training data and predicted on the test set. I used print statements to confirm the accuracy was 100% (for small test sets this is expected), and the probabilities sum to 1.



Part 2: Two binary classifiers vs. softmax

2.1 Implement TwoStageClassifier, reusing your BinaryLogisticRegression from Part 1

I reused the class from the binarylogisticregression, making sure to explicitly encode the labels correctly for the two-stage classification, using chain-rule. I verified this by fitting the data and predicting and checking the confusion matrix.

2.2 Implement softmax(S)

I reused the classes again for this, making sure to change the predict to use argmax, and using the given one_hot encoding class. I implemented softmax using the given equation.

2.3 Train both models

I repeated the basic training procedure and verified the accuracies, which involved a fix to the regularization.

<i>Model</i>	<i>Train Accuracy</i>	<i>Test Accuracy</i>	<i>Macro Precision</i>	<i>Macro Recall</i>	<i>Test log loss</i>
<i>Two-stage</i>	0.981 (103/105)	0.911 (41/45)	0.911	0.911	0.174
<i>Softmax</i>	0.981 (103/105)	0.933 (42/45)	0.935	0.933	0.192

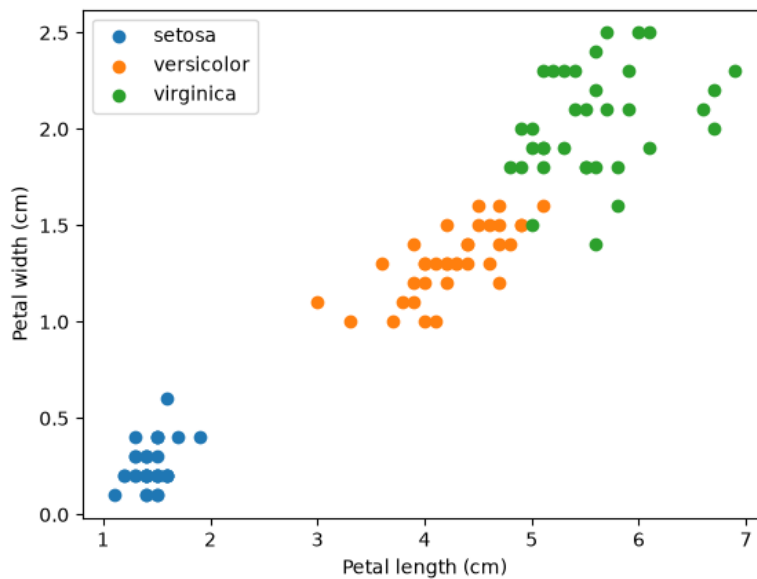
Part 3: Short-answer interpretation

3.1 Look at the two confusion matrices from Task 2.3

The Confusions between pairs is only versicolor vs virginica.

Two-stage: 2 versicolorvirginica and 2 virginicaversicolor (4/45 errors) and Softmax: 2 versicolor misclassified as virginica and 1 virginica misclassified as versicolor (3/45 errors).

Setosa: never contributed to an error in either model: all 15 test setosa are classified correctly and none of the other flowers is classed as predicted setosa (precision = recall = 1.0).



Scatter plot (Fig. 2): setosa appears as a distinct cluster at small petal lengths (1.9 cm) and widths (0.6 cm) in the training set, well separated from the other two, which is why stage 1 performs perfectly. Versicolor (length 3.0 - 5.1, width 1.0 - 1.6) and virginica (length 4.8 - 6.9 cm, width 1.4 - 2.5 cm) overlaps in a band where petal length is 4.8 - 5.1 cm and width is 1.4 - 1.8 cm. The prediction errors for the test flowers are in this part of the space (ex. petal length/width 4.8 / 1.8, 5.0 / 1.7 and 5.1 / 1.5 cm) for both models; 4.5 / 1.7 is an additional two-stage error), so a strict linear boundary cannot separate them perfectly.

3.2 In your own words, explain:

Why not $\max(q_1, q_2)$?

The two outputs ask different questions: q_1 is $P(\text{setosa} \mid x)$, and q_2 was trained only on non-setosa flowers, so it is $P(\text{second class} \mid x, \text{not setosa})$. Comparing the larger value means comparing that probabilities that live on different sample spaces, and their numbers need not add to 1 across the three classes. Also, a setosa flower is always removed before stage 2 so whatever the q_2 of the flower is fruitless. The chain rule $P(A \text{ and } B) = P(B \mid A)P(A)$ is what converts two of the options to a valid three-class distribution.

Role of $(1 - q_1)$:

This represents $P(\text{not setosa} \mid x)$, which is the probability that the stage-2 question is relevant. Multiplying the conditional q_1 (or $1 - q_2$) takes the conditional probability (determined given the case of q_1) and makes it unconditional (given all cases), $P(\text{versicolor} \mid x) = P(\text{not setosa} \mid x) * P(\text{versicolor} \mid \text{not setosa}, x)$. It also ensures that the three

probabilities sum to $q_1 + (1 - q_1) = 1$; if when $q_1 = 1$ all the other classes get close to zero probability regardless of what stage 2 says.

3.3 Recommendation

I would recommend softmax. Train accuracy is the same (0.981) but on the test set softmax is better (0.933 vs. 0.911; 42 versus 41 of 45 correct) with the higher macro precision/recall (0.935/0.933 versus 0.911/0.911) and it makes fewer virginica -> versicolor errors (1 vs. 2). It also discriminates all three classes simultaneously with a single model and a single loss, with no manually constructed conditional structure is used. Note that the gap is based on a single test flower, so it does not provide statistical significance.